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Multiple scales and averaging

Lecture content

3.1 Method of multiple scales

3.2 Methods of averaging

3.3 Motion of a simple pendulum

3.4 Motion of a current-carrying conductor

3.5 Motion of a particle on a rotating parabola

The conservative equation near equilibrium

Shift displacement to a stable equilibrium and expand the restoring acceleration. Through cubic order,

$$\ddot{x} + \alpha_1 x + \alpha_2 x^2 + \alpha_3 x^3 = 0, \qquad \omega_0 = \sqrt{\alpha_1}.$$

Here x is displacement from equilibrium; dots denote physical-time derivatives. $\alpha_1 > 0$ is linear stiffness per unit mass, α_2 is quadratic stiffness per unit mass, and α_3 is cubic stiffness per unit mass.

For displacement in metres, their units are s^{-2}, $\mathrm{m}^{-1}\mathrm{s}^{-2}$, and $\mathrm{m}^{-2}\mathrm{s}^{-2}$, respectively. For a spring, $\alpha_j = k_j/m$.

There is no damping or external force.

Why do we need equations for slow amplitude and phase?

A small frequency error can produce a large timing error after many cycles. Let R be the leading amplitude and $\Delta\omega$ a small nonlinear frequency correction:

$$x \approx R\cos[(\omega_0 + \Delta\omega)t] = R\cos[\omega_0 t + \beta(t)], \qquad \beta(t) = \Delta\omega\, t.$$

Expanding the cosine for small $\Delta\omega\, t$ gives

$$x \approx R\cos\omega_0 t - R\Delta\omega\, t\sin\omega_0 t + \cdots .$$

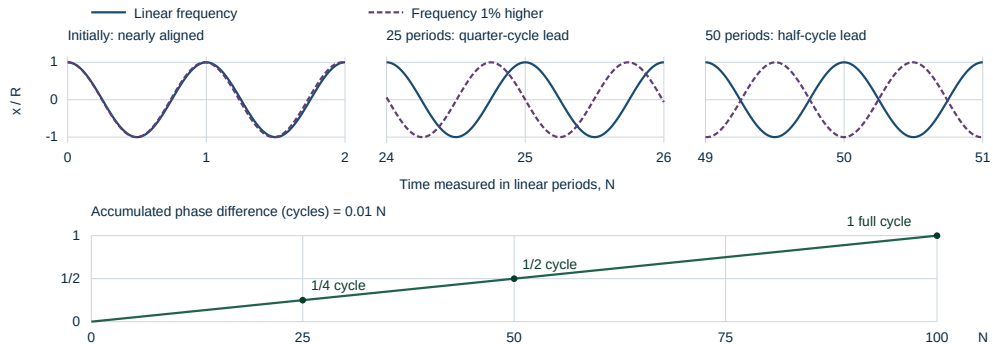
The term proportional to t grows even though the actual motion stays bounded. It represents accumulated phase error, not growing oscillation amplitude.

A 1% frequency correction accumulates 2π radians of phase difference over 100 linear periods. We therefore keep the phase inside the cosine and derive its slow evolution.

The amplitude and phase evolution equations are called the **slow flow**. In this conservative problem the leading amplitude is constant, but the phase relative to $\omega_0 t$ changes slowly. Separate time scales let us calculate that change without secular growth.

A 1% frequency difference accumulates over many cycles

$x/R = \cos\omega_0 t$ versus $\cos(1.01\omega_0 t)$; both amplitudes are constant. Linear cycle count: $N = \omega_0 t/(2\pi)$.



Slow time tracks the accumulated phase. At 100 periods the signals align again, but the faster oscillation has completed one extra cycle.

From accumulated phase to slow time

The plot shows that phase difference equals frequency difference multiplied by time: $\beta = \Delta\omega\, t$. Let $\varepsilon \ll 1$ count the small amplitude, and write a possible frequency expansion as

$$\omega = \omega_0 + \varepsilon\omega_1 + \varepsilon^2\omega_2 + \cdots,$$

where ω_1, ω_2 are frequency-correction coefficients. Then

$$\omega t = \omega_0 t + \omega_1(\varepsilon t) + \omega_2(\varepsilon^2 t) + \cdots = \omega_0 T_0 + \omega_1 T_1 + \omega_2 T_2 + \cdots.$$

This identifies $T_0 = t$, $T_1 = \varepsilon t$, and $T_2 = \varepsilon^2 t$. If ω_1 is comparable to ω_0, its phase contribution becomes order one when $\omega_0 t \sim \varepsilon^{-1}$; the second correction needs $\omega_0 t \sim \varepsilon^{-2}$.

The 1% plot illustrates either $\varepsilon = 0.01$ at first order or $\varepsilon = 0.1$ at second order, when the corresponding correction coefficient equals ω_0. The plot alone does not determine the perturbation order.

Which slow time matters for this oscillator?

Here ε counts amplitude: $x = \varepsilon x_1 + \cdots$. We keep both T_1 and T_2 initially and let the equations determine whether either produces a change.

The leading displacement is proportional to $\cos \omega_0 t$. Its square is

$$\cos^2 \omega_0 t = \tfrac{1}{2} + \tfrac{1}{2} \cos 2\omega_0 t.$$

Thus the quadratic term first produces a mean shift and a second harmonic, with no term at ω_0. The calculation will give no amplitude or phase evolution on T_1.

At the next order, the cubic term and the feedback from the quadratic correction do contain a term at ω_0. They produce the first frequency correction, of order ε^2, so the phase evolves on $T_2 = \varepsilon^2 t$.

These are different clocks for the same motion. During one fast cycle, T_1 advances only ε times as much as t, and T_2 only ε^2 times as much. The conservative leading amplitude remains constant.

Introduce the time scales and displacement expansion

Use a dimensionless small parameter ε to count powers of the small amplitude:

$$T_0 = t, \qquad T_1 = \varepsilon t, \qquad T_2 = \varepsilon^2 t.$$

$$x = \varepsilon x_1(T_0, T_1, T_2) + \varepsilon^2 x_2(T_0, T_1, T_2) + \varepsilon^3 x_3(T_0, T_1, T_2) + \cdots .$$

x_1, x_2, x_3 are successive displacement terms. The fast oscillation depends on T_0; changes that accumulate over longer times can depend on T_1 and T_2.

Here ε counts amplitude order; it is not a damping coefficient.

Apply the chain rule

Define the derivative notation $D_n = \partial/\partial T_n$. Since $\mathrm{d}T_n/\mathrm{d}t = \varepsilon^n$,

$$\frac{\mathrm{d}}{\mathrm{d}t} = D_0 + \varepsilon D_1 + \varepsilon^2 D_2 + \cdots .$$

Apply this derivative a second time and collect equal powers:

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2} = D_0^2 + 2\varepsilon D_0 D_1 + \varepsilon^2 (D_1^2 + 2D_0 D_2) + \cdots .$$

For example, $D_0 D_1 x_1 = \partial^2 x_1/(\partial T_0 \partial T_1)$.

Expand each term before collecting orders

$$
\begin{aligned}
\ddot{x} &= \varepsilon D_0^2 x_1 + \varepsilon^2 (D_0^2 x_2 + 2 D_0 D_1 x_1) \\
&\quad + \varepsilon^3 [D_0^2 x_3 + 2 D_0 D_1 x_2 + (D_1^2 + 2 D_0 D_2) x_1] + \cdots , \\
x^2 &= \varepsilon^2 x_1^2 + 2\varepsilon^3 x_1 x_2 + \cdots , \\
x^3 &= \varepsilon^3 x_1^3 + \cdots .
\end{aligned}
$$

Insert these expressions into $\ddot{x} + \omega_0^2 x + \alpha_2 x^2 + \alpha_3 x^3 = 0$. The coefficient of each retained power of ε must vanish.

Equations at the three retained orders

$$
\begin{aligned}
O(\varepsilon): \quad & D_0^2 x_1 + \omega_0^2 x_1 = 0, \\
O(\varepsilon^2): \quad & D_0^2 x_2 + \omega_0^2 x_2 = -2 D_0 D_1 x_1 - \alpha_2 x_1^2, \\
O(\varepsilon^3): \quad & D_0^2 x_3 + \omega_0^2 x_3 = -2 D_0 D_1 x_2 - D_1^2 x_1 \\
& \qquad\qquad - 2 D_0 D_2 x_1 - 2\alpha_2 x_1 x_2 - \alpha_3 x_1^3.
\end{aligned}
$$

Quadratic stiffness first appears in x_2; cubic stiffness first appears in x_3.

Write the leading harmonic in complex form

The solution of $D_0^2 x_1 + \omega_0^2 x_1 = 0$ is written

$$x_1 = A(T_1, T_2)e^{i\omega_0 T_0} + \bar{A}(T_1, T_2)e^{-i\omega_0 T_0}.$$

$i^2 = -1$ and $\bar{A}$ is the complex conjugate of A. The two terms add to a real displacement.

With $A = \frac{1}{2}ae^{i\beta}$, Euler's identity gives

$$x_1 = \tfrac{1}{2}a[e^{i(\omega_0 T_0 + \beta)} + e^{-i(\omega_0 T_0 + \beta)}] = a\cos(\omega_0 T_0 + \beta).$$

a is the leading amplitude coefficient and β the phase.

Second order: identify the resonant term

Squaring the leading harmonic gives

$$x_1^2 = A^2 e^{2i\omega_0 T_0} + 2A\bar{A} + \bar{A}^2 e^{-2i\omega_0 T_0}.$$

Therefore the second-order equation contains

$$D_0^2 x_2 + \omega_0^2 x_2 = -2i\omega_0 D_1 A\, e^{i\omega_0 T_0} - \alpha_2 A^2 e^{2i\omega_0 T_0} - \alpha_2 A\bar{A} + \text{cc}.$$

Here cc means add the complex conjugate of the displayed right-hand terms. The term at ω_0 is resonant; the mean and second harmonic are not.

$$\boxed{D_1 A = 0.}$$

The amplitude has no dependence on T_1.

Solve the mean and second-harmonic terms

After $D_1 A = 0$, try $x_2 = Be^{2i\omega_0 T_0} + \bar{B}e^{-2i\omega_0 T_0} + C$.

For the second harmonic, differentiation gives $(-4\omega_0^2 + \omega_0^2)B = -\alpha_2 A^2$. For the constant term, $\omega_0^2 C = -2\alpha_2 A\bar{A}$. Hence

$$\boxed{x_2 = \frac{\alpha_2 A^2}{3\omega_0^2} e^{2i\omega_0 T_0} + \frac{\alpha_2 \bar{A}^2}{3\omega_0^2} e^{-2i\omega_0 T_0} - \frac{2\alpha_2 A\bar{A}}{\omega_0^2}.}$$

Free homogeneous terms are absorbed into the amplitude and phase constants, which are fitted to the initial data.

Third order: calculate the fundamental coefficient

Since $D_1 A = 0$, the terms involving D_1 vanish. Only the coefficient of $e^{i\omega_0 T_0}$ is needed for solvability.

$$[x_1 x_2]_{\omega_0} = \left(-2 + \frac{1}{3}\right) \frac{\alpha_2}{\omega_0^2} A^2 \bar{A} = -\frac{5\alpha_2}{3\omega_0^2} A^2 \bar{A},$$

$$[x_1^3]_{\omega_0} = 3A^2 \bar{A}.$$

Thus $-2\alpha_2 x_1 x_2 - \alpha_3 x_1^3$ contributes $(10\alpha_2^2/(3\omega_0^2) - 3\alpha_3) A^2 \bar{A}$ at the fundamental.

This is where the squared quadratic coefficient enters the frequency correction.

Third-order solvability condition

The resonant coefficient of the x_3 equation is

$$-2i\omega_0 D_2 A + \left(\frac{10\alpha_2^2}{3\omega_0^2} - 3\alpha_3 \right) A^2 \bar{A}.$$

Set it to zero so the correction does not contain a secular term:

$$\boxed{2i\omega_0 D_2 A + \frac{9\alpha_3\omega_0^2 - 10\alpha_2^2}{3\omega_0^2} A^2 \bar{A} = 0.}$$

Both nonlinearities contribute, even though the quadratic term had no fundamental harmonic at second order.

Separate amplitude and phase

Set $A = \frac{1}{2}a(T_2)e^{i\beta(T_2)}$. A prime on this slide means $\mathrm{d}/\mathrm{d}T_2$:

$$D_2 A = \tfrac{1}{2}(a' + ia\beta')e^{i\beta}, \qquad A^2\bar{A} = \tfrac{1}{8}a^3 e^{i\beta}.$$

Substitute into the solvability equation and divide by $e^{i\beta}$:

$$i\omega_0 a' - \omega_0 a\beta' + \frac{9\alpha_3\omega_0^2 - 10\alpha_2^2}{24\omega_0^2}a^3 = 0.$$

$$\boxed{a' = 0, \qquad \beta' = \frac{9\alpha_3\omega_0^2 - 10\alpha_2^2}{24\omega_0^3}a^2.}$$

The conservative amplitude is constant; the phase accumulates on T_2.

Recover frequency and displacement

Integrate β' and use $T_2 = \varepsilon^2 t$. Combining the phase correction with $\omega_0 t$ gives

$$\boxed{\omega = \omega_0 + \left(\frac{3\alpha_3}{8\omega_0} - \frac{5\alpha_2^2}{12\omega_0^3} \right) \varepsilon^2 a^2 + O(\varepsilon^3).}$$

Using $A = \frac{1}{2} a e^{i\beta}$ in x_2, the displacement is

$$x = \varepsilon a \cos(\omega t + \beta_0) - \frac{\varepsilon^2 a^2 \alpha_2}{2\omega_0^2} \left[1 - \frac{1}{3} \cos(2\omega t + 2\beta_0) \right] + O(\varepsilon^3).$$

The quadratic term produces a mean shift and second harmonic.

Fit the constants to a release from rest

The physical leading harmonic amplitude is εa. It is not automatically the positive release displacement R.

For release at the positive turning point, choose $\beta_0 = 0$. At $t = 0$, the displacement formula gives

$$R = \varepsilon a - \frac{\alpha_2}{3\omega_0^2}(\varepsilon a)^2 + O(\varepsilon^3).$$

Solve iteratively, replacing εa by R inside the quadratic correction:

$$\boxed{\varepsilon a = R + \frac{\alpha_2 R^2}{3\omega_0^2} + O(R^3).}$$

The sine terms vanish at $t = 0$, so the retained waveform starts at rest. To the leading R^2 frequency correction, $(\varepsilon a)^2$ can be replaced by R^2.

Transform displacement into amplitude and phase

Starting from the same conservative equation, use variation of parameters:

$$x = a(t)\cos\phi, \qquad \dot{x} = -\omega_0 a(t)\sin\phi, \qquad \phi = \omega_0 t + \beta(t).$$

Here $a(t)$ is the displacement amplitude coordinate and $\beta(t)$ is phase relative to the linear oscillation. In this section, dots again mean $\mathrm{d}/\mathrm{d}t$.

Differentiate the first expression and equate it to the specified velocity:

$$\boxed{\dot{a}\cos\phi - a\dot{\beta}\sin\phi = 0.}$$

This relation ensures that the two amplitude–phase expressions describe the same motion.

Derive the second amplitude–phase equation

Differentiate the velocity expression:

$$\ddot{x} = -\omega_0 \dot{a} \sin\phi - \omega_0 a(\omega_0 + \dot{\beta}) \cos\phi.$$

Insert it into $\ddot{x} + \omega_0^2 x + \alpha_2 x^2 + \alpha_3 x^3 = 0$. The linear terms cancel:

$$\boxed{\dot{a} \sin\phi + a\dot{\beta} \cos\phi = \frac{\alpha_2 a^2 \cos^2\phi + \alpha_3 a^3 \cos^3\phi}{\omega_0}.}$$

Together with $\dot{a} \cos\phi - a\dot{\beta} \sin\phi = 0$, this is a pair of linear equations for $\dot{a}$ and $a\dot{\beta}$.

Solve for the exact instantaneous rates

Multiply the first equation by $\cos\phi$ and the second by $\sin\phi$, then add. For the phase rate, multiply the second by $\cos\phi$ and subtract the first times $\sin\phi$.

$$\boxed{\dot{a} = \frac{\sin\phi}{\omega_0}(\alpha_2 a^2 \cos^2\phi + \alpha_3 a^3 \cos^3\phi),}$$

$$\boxed{\dot{\beta} = \frac{\cos\phi}{\omega_0}(\alpha_2 a \cos^2\phi + \alpha_3 a^2 \cos^3\phi).}$$

These exact amplitude–phase equations follow from the original displacement equation. They still contain variations within each oscillation.

Evaluate the cycle means

For the first approximation, hold a and β fixed over one cycle and define

$$\langle g \rangle = \frac{1}{2\pi} \int_0^{2\pi} g(\phi)\, \mathrm{d}\phi.$$

Integrals of $\sin\phi\cos^2\phi$ and $\sin\phi\cos^3\phi$ vanish because they are derivatives of periodic powers of $\cos\phi$. Also,

$$\langle \cos^3\phi \rangle = 0, \qquad \cos^4\phi = \frac{3 + 4\cos 2\phi + \cos 4\phi}{8} \quad \Longrightarrow \quad \langle \cos^4\phi \rangle = \frac{3}{8}.$$

Only the cubic term has a nonzero mean in the first-order phase equation.

First-order averaging result and its limitation

Applying the cycle means gives the first approximation:

$$\boxed{\dot{a} = 0, \qquad \dot{\beta} = \frac{3\alpha_3 a^2}{8\omega_0}, \qquad \omega \approx \omega_0 + \frac{3\alpha_3 a^2}{8\omega_0}.}$$

Compare with the multiple-scales result, expressed in physical leading amplitude a:

$$\omega \approx \omega_0 + \left(\frac{3\alpha_3}{8\omega_0} - \frac{5\alpha_2^2}{12\omega_0^3} \right) a^2.$$

For $\alpha_2 = 0$, the retained corrections agree. When $\alpha_2 \neq 0$, first-order averaging misses the quadratic feedback; a higher-order averaging calculation is required.

Why averaging a guessed frequency fails

Treating $\sqrt{\alpha_1 + \alpha_2 x + \alpha_3 x^2}$ as an instantaneous frequency and simply averaging it after $x = a\cos\phi$ gives an incorrect correction.

That operation gives

$$\omega_{\text{guessed}} = \omega_0 + \left(\frac{\alpha_3}{4\omega_0} - \frac{\alpha_2^2}{16\omega_0^3} \right) a^2 + \cdots ,$$

which differs from the consistently derived correction

$$\omega = \omega_0 + \left(\frac{3\alpha_3}{8\omega_0} - \frac{5\alpha_2^2}{12\omega_0^3} \right) a^2 + \cdots .$$

Average the derived amplitude–phase equations.

Cubic spring: apply both methods

For a mass m with restoring force $kx + k_3x^3$, divide $m\ddot{x} + kx + k_3x^3 = 0$ by m:

$$\alpha_1 = k/m, \qquad \alpha_2 = 0, \qquad \alpha_3 = k_3/m.$$

Take $m = 1$ kg, $k = 100$ N/m, cubic stiffness $k_3 = 1000$ N/m^3, and release $R = 0.10$ m. Then

$$\omega_0 = 10 \text{ rad/s}, \qquad \omega \approx 10 + \frac{3(1000)(0.10)^2}{8(10)} = 10.375 \text{ rad/s}.$$

Multiple scales and first-order averaging give the same leading frequency here because quadratic stiffness is absent.

Cubic spring: enforce the release in the waveform

For $\alpha_2 = 0$, set $\omega = \omega_0 + 3\alpha_3 R^2/(8\omega_0)$ and try

$$x = R\cos\omega t + B(\cos 3\omega t - \cos\omega t).$$

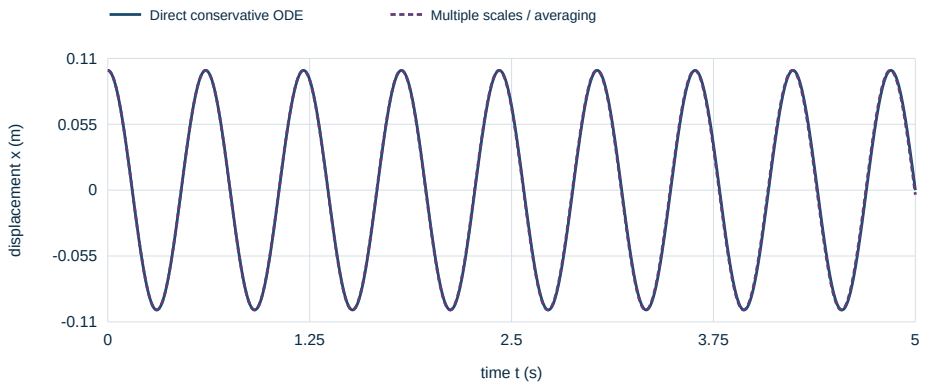
The two B terms cancel at release. At the retained order, the third-harmonic balance is $-8\omega_0^2 B + \alpha_3 R^3/4 = 0$, so

$$B = \frac{\alpha_3 R^3}{32\omega_0^2} = \frac{1000(0.10)^3}{32(100)} = 0.0003125 \text{ m}.$$

$$x \approx 0.10\cos(10.375t) + 0.0003125[\cos(31.125t) - \cos(10.375t)] \text{ m}.$$

$x(0) = 0.10$ m and $\dot{x}(0) = 0$. This is the pure-cubic specialization of the third-order expansion.

Cubic spring: compare with direct integration



Quadratic and cubic spring: compare the predictions

Use $m = 1$ kg, $k = 100$ N/m, quadratic stiffness $k_2 = 200$ N/m^2, cubic stiffness $k_3 = 100$ N/m^3, and $R = 0.050$ m.

$$\omega_0 = 10, \qquad \alpha_2 = 200, \qquad \alpha_3 = 100.$$

The leading release-amplitude frequency correction is

$$\omega \approx 10 + \left[\frac{3(100)}{8(10)} - \frac{5(200)^2}{12(10)^3}\right](0.050)^2 = 9.96771 \text{ rad/s}.$$

First-order averaging keeps only the cubic part and gives $10 + 3(100)(0.050)^2/[8(10)] = 10.009375$ rad/s. The quadratic feedback reverses the predicted direction of the frequency shift in this example.

Check the mixed spring against the energy integral

For this conservative equation, the potential per unit mass is

$$U(x) = \tfrac{1}{2}\omega_0^2 x^2 + \tfrac{1}{3}\alpha_2 x^3 + \tfrac{1}{4}\alpha_3 x^4.$$

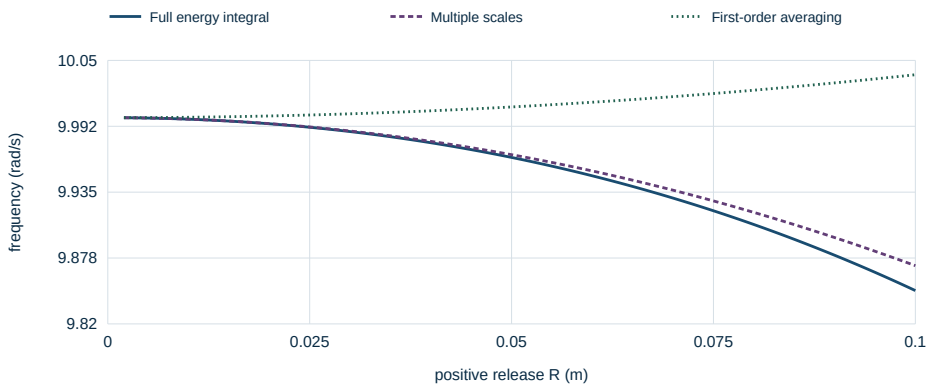
At release $E = U(R)$. Find the negative turning point x_- from $U(x_-) = E$; then

$$T = 2\int_{x_-}^{R} \frac{\mathrm{d}x}{\sqrt{2[E - U(x)]}}, \qquad \omega = \frac{2\pi}{T}.$$

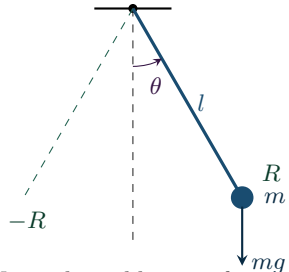
At $R = 0.050$ m, numerical evaluation gives $\omega = 9.96531$ rad/s. Multiple scales gives 9.96771 rad/s; first-order averaging gives 10.00938 rad/s. The next plot repeats this calculation over a range of releases.

The energy integral retains the full quadratic and cubic restoring terms.

Mixed spring: the quadratic contribution matters



Pendulum: physical model and release



Vertical equilibrium: $\theta = 0$

A point mass m hangs from a massless rod of length l. The pivot is fixed; friction is neglected.

Release at $\theta(0) = R$, $\dot{\theta}(0) = 0$. Angles are in radians.

Gravity produces torque $-mgl\sin\theta$ about the pivot:

$$ml^2\ddot{\theta} = -mgl\sin\theta.$$

$$\ddot{\theta} + \frac{g}{l}\sin\theta = 0.$$

How does increasing the release angle change the period?

Simple pendulum: identify the coefficients

The pendulum equation is $\ddot{\theta} + \omega_0^2 \sin\theta = 0$, with $\omega_0^2 = g/l$.

For a small angle in radians, expand $\sin\theta = \theta - \theta^3/6 + \cdots$. Therefore, in the general conservative equation,

$$\alpha_1 = \omega_0^2, \qquad \alpha_2 = 0, \qquad \alpha_3 = -\omega_0^2/6.$$

With angular release amplitude R, insert these into either leading frequency correction:

$$\boxed{\omega \approx \omega_0 \left(1 - \frac{R^2}{16}\right), \qquad T \approx \frac{2\pi}{\omega_0}\left(1 + \frac{R^2}{16}\right).}$$

The period formula is truncated consistently through R^2.

Simple pendulum: numerical substitution

Let $l = 1.00$ m, $g = 9.81$ m/s^2 and $R = 30^\circ = \pi/6 = 0.52360$ rad.

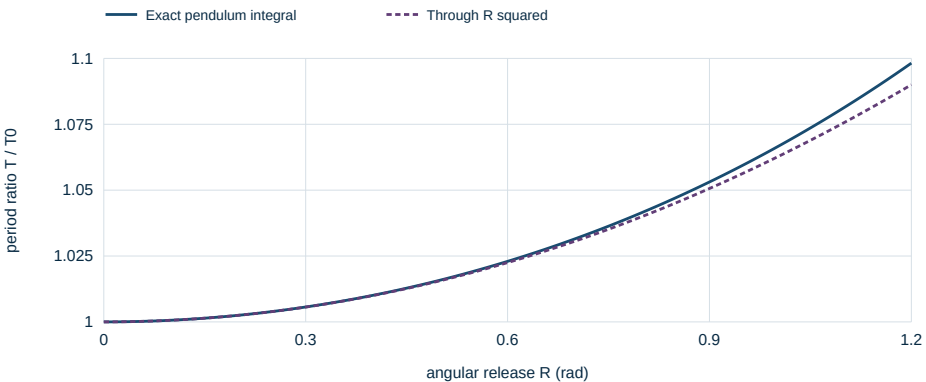
$$\omega_0 = \sqrt{9.81} = 3.13209 \text{ rad/s}, \qquad \frac{R^2}{16} = 0.0171347.$$

$$\omega \approx 3.13209(1 - 0.0171347) = 3.07842 \text{ rad/s}.$$

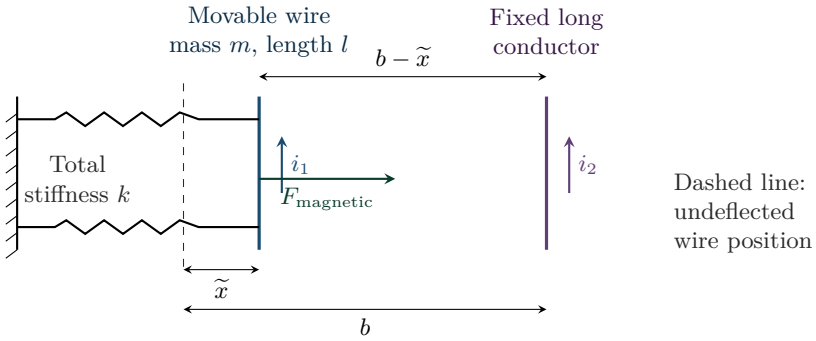
$$T \approx \frac{2\pi}{3.13209}(1 + 0.0171347) = 2.04044 \text{ s}.$$

The nonlinear pendulum has a lower frequency and longer period than its linear approximation. The next plot compares this expansion with the exact pendulum period.

Pendulum: compare with the exact period



Conductor: geometry, currents and displacement



Parallel currents attract. Positive displacement reduces the gap and strengthens the attraction. The springs oppose that displacement.

Current-carrying conductor: spring versus attraction

A movable wire of mass m and length l is restrained by springs of total stiffness k. A parallel fixed conductor is a distance b from its undeflected position. Displacement $\widetilde{x}$ toward the fixed conductor reduces the gap to $b - \widetilde{x}$.

For parallel currents i_1, i_2 in air, the attractive force in SI units is

$$F_{\text{magnetic}} = \frac{\mu_0 i_1 i_2 l}{2\pi(b - \widetilde{x})}, \qquad m\frac{\mathrm{d}^2\widetilde{x}}{\mathrm{d}\widetilde{t}^2} + k\widetilde{x} - F_{\text{magnetic}} = 0.$$

Here μ_0 is the magnetic permeability of free space; $\widetilde{t}$ is physical time.

The spring force grows with displacement, but the magnetic attraction grows rapidly as the gap closes. Can the spring still hold a stable equilibrium?

Conductor: reduce the model to one parameter

Define dimensionless displacement and time:

$$x = \frac{\widetilde{x}}{b}, \qquad t = \sqrt{\frac{k}{m}}\,\widetilde{t}, \qquad \Lambda = \frac{\mu_0 i_1 i_2 l}{2\pi k b^2}.$$

Since $\mathrm{d}^2\widetilde{x}/\mathrm{d}\widetilde{t}^2 = b(k/m)\ddot{x}$, division of the force equation by kb gives

$$\boxed{\ddot{x} + x - \frac{\Lambda}{1-x} = 0, \qquad x < 1.}$$

Here dots denote dimensionless-time derivatives. Λ measures magnetic attraction relative to spring stiffness.

Conductor: locate the stable and unstable equilibria

At equilibrium $\dot{x} = \ddot{x} = 0$, so

$$x - \frac{\Lambda}{1-x} = 0 \implies x(1-x) = \Lambda \implies x^2 - x + \Lambda = 0.$$

$$x_1 = \frac{1}{2} + \sqrt{\frac{1}{4} - \Lambda}, \qquad x_2 = \frac{1}{2} - \sqrt{\frac{1}{4} - \Lambda}.$$

For $0 < \Lambda < 1/4$, expand about an equilibrium: $x = x_i + u$ gives

$$\ddot{u} + \left[1 - \frac{\Lambda}{(1-x_i)^2}\right] u \approx 0.$$

At x_2 the bracket is positive: a center with small oscillations. At x_1 it is negative: an unstable saddle. At $\Lambda = 1/4$ they merge; above it no equilibrium remains.

Conductor: energy determines whether motion is bounded

Multiply the equation by $\dot{x}$ and integrate in time:

$$\dot{x}\ddot{x} + \left(x - \frac{\Lambda}{1-x}\right)\dot{x} = 0 \quad \Longrightarrow \quad \frac{1}{2}\dot{x}^2 + F(x) = h,$$

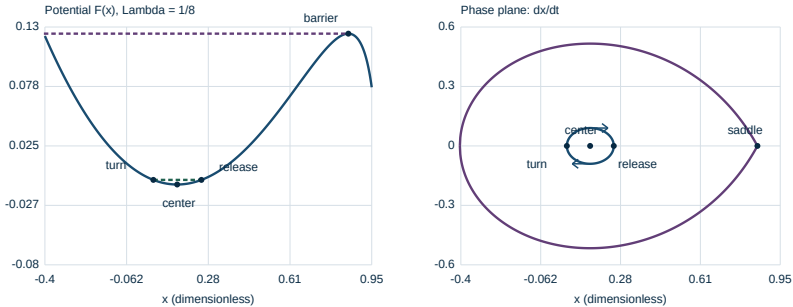
$$F(x) = \frac{1}{2}x^2 + \Lambda \ln(1-x).$$

Here F is dimensionless potential and h is the conserved energy level.

Release at $x = x_2 + R$ gives $h = F(x_2 + R)$. The turning points satisfy $F(x) = h$, and the phase curve is $\dot{x} = \pm\sqrt{2[h - F(x)]}$.

A release in the well with $h < F(x_1)$ gives bounded oscillations. The saddle energy is the barrier; energy alone is insufficient unless the initial position is also on the well side.

Conductor: the well and its closed phase curve



$\Lambda = 1/8$: $x_2 = 0.14645$, $x_1 = 0.85355$. Blue: release $x = x_2 + 0.10$; purple: the limiting curve through the saddle. Arrows show time direction.

Conductor: expand about the displaced equilibrium

Put $x = x_2 + u$, where u is displacement from the stable equilibrium. Expand the reciprocal for $|u| < 1 - x_2$:

$$\frac{1}{1 - x_2 - u} = \frac{1}{1 - x_2}\left[1 + \frac{u}{1 - x_2} + \frac{u^2}{(1 - x_2)^2} + \frac{u^3}{(1 - x_2)^3} + \cdots\right].$$

The constant terms cancel because $x_2 = \Lambda/(1 - x_2)$. The remaining equation is

$$\ddot{u} + \alpha_1 u + \alpha_2 u^2 + \alpha_3 u^3 + \cdots = 0,$$

$$\alpha_1 = 1 - \frac{\Lambda}{(1 - x_2)^2}, \quad \alpha_2 = -\frac{\Lambda}{(1 - x_2)^3}, \quad \alpha_3 = -\frac{\Lambda}{(1 - x_2)^4}.$$

This is the same local equation used in the multiple-scales calculation. Both nonlinear contributions lower the leading frequency.

Conductor: calculate and check the frequency

For $\Lambda = 1/8$ and release $u(0) = R = 0.10$, $\dot{u}(0) = 0$:

$$\alpha_1 = 0.828427, \quad \alpha_2 = -0.201010, \quad \alpha_3 = -0.235498, \quad \omega_0 = \sqrt{0.828427} = 0.910180.$$

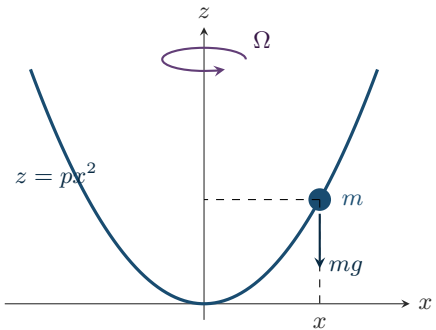
$$\omega \approx 0.910180 + \left[\frac{3(-0.235498)}{8(0.910180)} - \frac{5(-0.201010)^2}{12(0.910180)^3}\right](0.10)^2 = 0.908986.$$

For a full-model check, use $\mathrm{d}t = \mathrm{d}x/\sqrt{2[h - F(x)]}$ on the positive-velocity half-cycle:

$$T = 2\int_{0.048056}^{0.246447} \frac{\mathrm{d}x}{\sqrt{2[h - F(x)]}} = 6.91230, \qquad \frac{2\pi}{T} = 0.908986.$$

Time and frequency here are dimensionless. Multiply ω by $\sqrt{k/m}$ for rad/s. The approximation is local to x_2, not valid near loss of equilibrium.

Rotating parabola: geometry and constraints



The rigid wire rotates about the vertical z axis at prescribed constant speed Ω.

The bead slides without friction. The constraint $z = px^2$ leaves one coordinate, x, measured in the rotating plane of the wire.

p sets the curvature (units m^{-1}). Release at $x(0) = R$, $\dot{x}(0) = 0$.

Will the bead return toward the vertex, or move outward as the rotation speed increases?

Rotating parabola: gravity competes with rotation

A bead of mass m slides without friction on the wire $z = px^2$, rotating about the vertical axis at constant angular speed Ω. Here $p > 0$ has units m^{-1}; x is the signed horizontal coordinate along the rotating wire.

Gravity draws the bead toward the vertex; rotation tends to move it outward. Its three velocity components have magnitudes $|\dot{x}|$, $|\Omega x|$, and $|\dot{z}|$, with $\dot{z} = 2px\dot{x}$.

$$T = \tfrac{1}{2}m\left[\dot{x}^2 + \Omega^2 x^2 + \dot{z}^2\right] = \tfrac{1}{2}m\left[(1 + 4p^2x^2)\dot{x}^2 + \Omega^2 x^2\right],$$

$$V = mgz = mgpx^2.$$

Time is physical time again. The factor $1 + 4p^2x^2$ accounts for the additional vertical motion as the bead travels along the curved wire.

Rotating parabola: derive the equation of motion

Form $L = T - V$ and apply $\mathrm{d}(\partial L/\partial \dot{x})/\mathrm{d}t - \partial L/\partial x = 0$:

$$\frac{\partial L}{\partial \dot{x}} = m(1 + 4p^2x^2)\dot{x},$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{x}} = m\left[(1 + 4p^2x^2)\ddot{x} + 8p^2x\dot{x}^2\right],$$

$$\frac{\partial L}{\partial x} = 4mp^2x\dot{x}^2 + m\Omega^2 x - 2mgpx.$$

Subtract and divide by m. Define $\Lambda = 2gp - \Omega^2$, with units s^{-2}:

$$\boxed{(1 + 4p^2x^2)\ddot{x} + 4p^2x\dot{x}^2 + \Lambda x = 0.}$$

Λ now measures gravity minus rotation and has units s^{-2}.

Rotating parabola: the conserved quantity

Multiply the equation by $2\dot{x}$. The product rule gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[(1+4p^2x^2)\dot{x}^2\right] = 2(1+4p^2x^2)\dot{x}\ddot{x} + 8p^2x\dot{x}^3.$$

Therefore

$$\boxed{(1+4p^2x^2)\dot{x}^2 + \Lambda x^2 = h, \qquad \dot{x}^2 = \frac{h-\Lambda x^2}{1+4p^2x^2}.}$$

Release at $x = R$ with zero velocity gives $h = \Lambda R^2$. For $\Lambda > 0$, motion is bounded between $-R$ and R. This is the Jacobi integral. The physical energy $T + V$ need not be constant: maintaining the wire's rotation can exchange energy with the bead.

Rotating parabola: when do oscillations exist?

Near the vertex, the equation reduces to $\ddot{x} + \Lambda x \approx 0$.

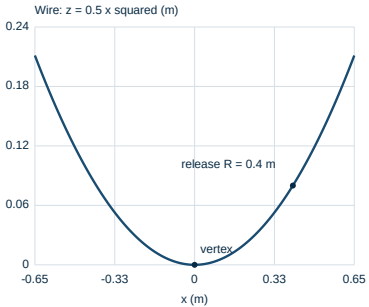
$\Omega^2 < 2gp$	$\Lambda > 0$: the vertex is a center; nearby motions oscillate.
$\Omega^2 = 2gp$	$\Lambda = 0$: every position at rest is an equilibrium; there is no restoring oscillation.
$\Omega^2 > 2gp$	$\Lambda < 0$: the vertex is an unstable saddle; a bead released away from it moves outward.

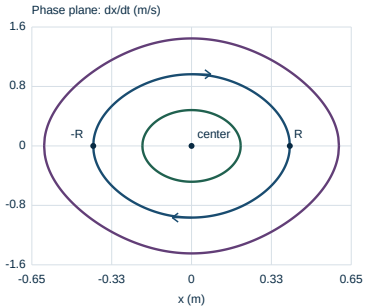
For $p = 0.50\ \mathrm{m}^{-1}$ and $g = 9.81\ \mathrm{m/s}^2$,

$$\Omega_{\text{critical}} = \sqrt{2gp} = \sqrt{9.81} = 3.13209\ \text{rad/s}.$$

Choose $\Omega = 2.00$ rad/s: $\Lambda = 9.81 - 4 = 5.81\ \mathrm{s}^{-2} > 0$. We next calculate the period of this bounded motion.

Rotating parabola: geometry and phase motion





$p = 0.50\ \mathrm{m}^{-1}$, $\Omega = 2.00$ rad/s. Closed phase curves correspond to releases $R = 0.2$, 0.4, 0.6 m. The curve shape alone does not determine the period.

Rotating parabola: obtain the period from the motion

For $\Lambda > 0$, the Jacobi integral gives the positive speed between 0 and R:

$$\dot{x} = \sqrt{\frac{\Lambda(R^2 - x^2)}{1 + 4p^2x^2}}, \qquad \frac{T}{4} = \frac{1}{\sqrt{\Lambda}} \int_0^R \frac{\sqrt{1 + 4p^2x^2}}{\sqrt{R^2 - x^2}}\, \mathrm{d}x.$$

Set $x = R\sin\theta$: $\mathrm{d}x = R\cos\theta\,\mathrm{d}\theta$ and $\sqrt{R^2 - x^2} = R\cos\theta$. These factors cancel:

$$\boxed{T = \frac{4}{\sqrt{\Lambda}} \int_0^{\pi/2} \sqrt{1 + 4p^2R^2\sin^2\theta}\,\mathrm{d}\theta.}$$

As R increases, the integrand increases. Thus the period increases even though the effective potential in the Jacobi integral is quadratic.

Rotating parabola: expand and evaluate the period

For $pR \ll 1$, use $\sqrt{1+q} = 1 + q/2 + O(q^2)$ inside the integral:

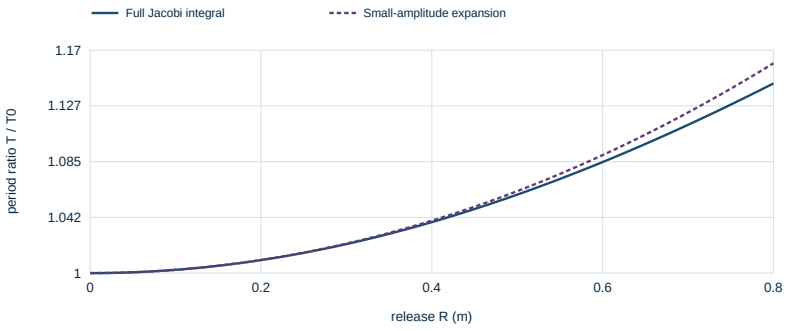
$$T \approx \frac{4}{\sqrt{\Lambda}} \left[\int_0^{\pi/2} 1 \, \mathrm{d}\theta + 2p^2 R^2 \int_0^{\pi/2} \sin^2\theta \, \mathrm{d}\theta \right] = \frac{2\pi}{\sqrt{\Lambda}} (1 + p^2 R^2).$$

With $p = 0.50$ m^{-1}, $\Lambda = 5.81$ s^{-2}, $R = 0.40$ m:

$$p^2 R^2 = (0.50)^2 (0.40)^2 = 0.0400, \qquad T \approx \frac{2\pi}{\sqrt{5.81}} (1.04) = 2.71097 \text{ s}.$$

The full integral gives $T = 2.70804$ s, so the approximation is about 0.108% high. The leading frequency is $\omega \approx \sqrt{\Lambda}(1 - p^2 R^2)$.

Rotating parabola: amplitude increases the period



$T_0 = 2\pi/\sqrt{\Lambda}$, $p = 0.50$ m^{-1}. The approximation agrees near $R = 0$; its error grows as the neglected powers of pR become significant.

Sources for the methods and examples

[1] A. H. Nayfeh and D. T. Mook
Nonlinear Oscillations. Wiley, 1979.

Secs. 2.3.3 and 2.3.5: multiple scales and averaging.
Secs. 2.4.1–2.4.3: simple pendulum, current-carrying conductor, and particle on a rotating parabola.